

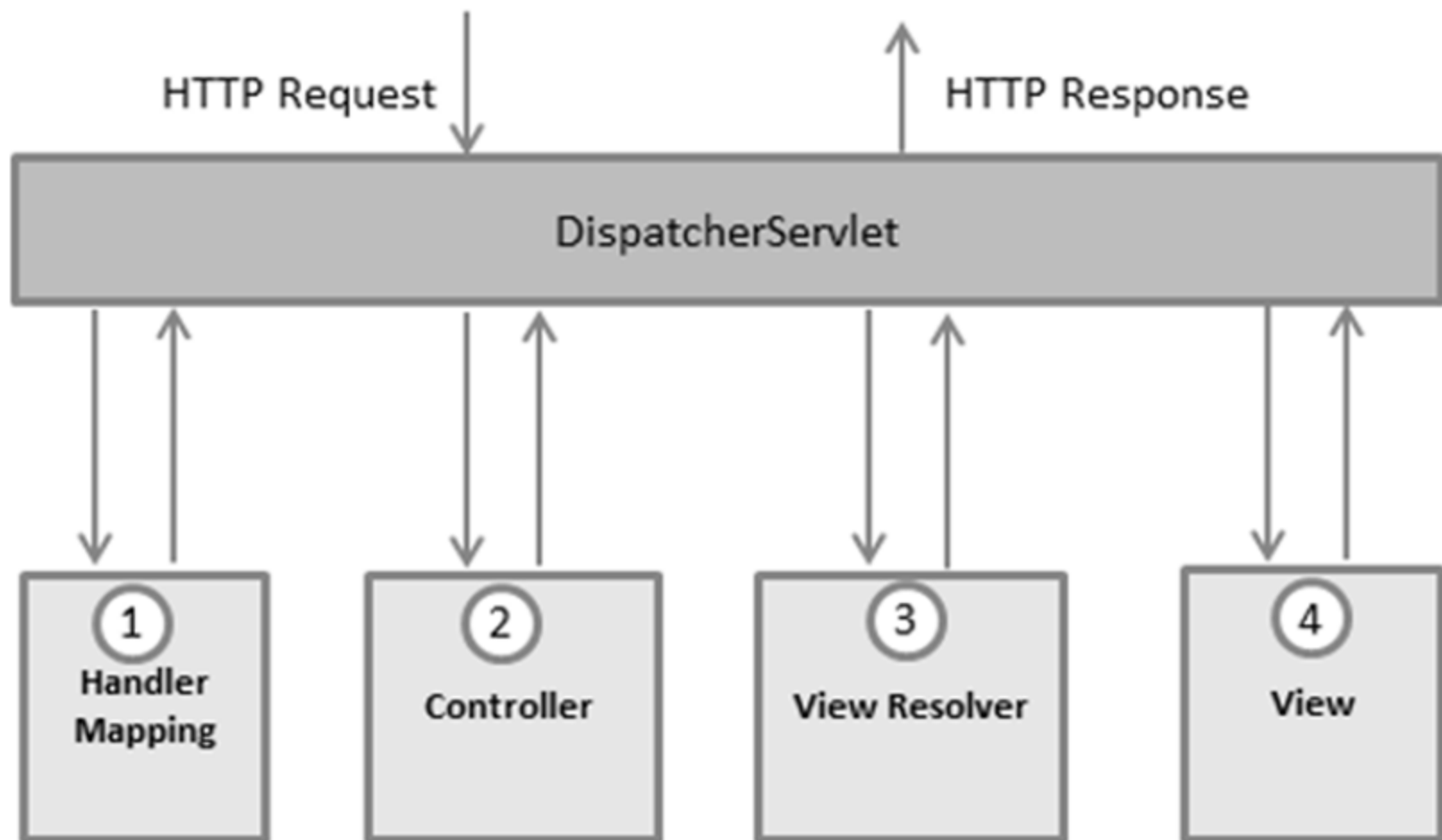


Spring MVC

1. What is MVC
2. Components and flow
3. Front controller - DispatcherServlet
4. Steps

- The MVC pattern results in separating the different aspects of the application (input logic, business logic, and UI logic), while providing a loose coupling between these elements.
- The Spring Web MVC framework provides ready components that can be used to implement the given requirement.
- Benefits Of Spring Web MVC :
 - Easy to learn
 - it gives developer flexibility to implement or integrate any other design pattern
 - Spring provides a custom a tag library
 - Project setup with Spring is quite easy and almost all classes are POJO classes.
 - Supports Annotation based programming along with XML which is faster and easier

2. Components and flow



- All requests will be sent to Front Controller (Dispatcher Servlet)
- Front Controller finds controller using handler mapping beans.
- Controller that took the request, processes the request, by sending it to suitable service class.
- After all processing is done, Controller receives the model from the Service or Data Access layer.
- Controller will return view name
- Dispatcher servlet finds the view template, using view resolver
- Using View template, model and view page is build and sent back to the Front Controller.
- Front controller sends the constructed view page to the browser to render it for the user requested.

- Front Controller is defined as “a controller that handles all requests for a Web site”.
- It intercepts every request and then dispatches requests to an appropriate controller.
- DispatcherServlet needs to be configured in deployment descriptor i.e. web.xml using <servlet> and <servlet-mapping> tags.
- Upon initialization of a DispatcherServlet, the framework looks for a file named [servlet-name]-servlet.xml in the WEB-INF directory of your web application.

- Using Eclipse IDE, create a Maven Project by selecting Web Archetype.
- Update Pom.xml to add the dependency for spring-webmvc module
- Update the web.xml to map DispatcherServlet as front controller of the web application using `<servlet>` and `<servlet-mapping>` tags.
- Create `<name of DispatcherServlet>-servlet.xml` under the `src/main/webapp/WEB-INF` folder and define the beans.
- Create controller classes.
- Create service classes and required POJOs
- Create views mainly JSP files